ZOO501 CURRENT FINAL TERM MCQS

1. Due to **Loss** in hydrogen ions, pH rises.
2. As a result of gastrulation, ectoderm, mesoderm and **Endoderm** are formed.
3. Male leopard **Frogs** make their sperm in summer.
4. **Differential gene transcription** is step for gene expression regulation.
5. Coding regions of DNA **exon.**
6. How many domains are present for transcription **3**
7. pdx1 is present in **Pancreatic** And Duodenal1 gene.
8. Bacteria transmit genes from one cell to other by **sex pili.**
9. Cells rearrange their **DNA** to create new immunoglobin and antigen receptor gene.
10. Sea urchins contain **25,000 to 50,000** mRNA.
11. In many species, sperm are attracted towards eggs by **chemotaxis**.
12. Acrosomal protein mediating recognition by protein **bindin.**
13. Sea water has **High** sodium ion conc.
14. Eggs have sparse **equally** spaced yolk are icolecithel.
15. Cytoplasm of blastodisc become embryo this meroblastic division called **discoidal.**
16. **The micromere** constitute major signaling center.
17. Osteoblasts secrete collagen proteoglycan **matrix** that is able to bind calcium salts
18. Tgf beta family secreted from the surface **ectoderm** and roof of neural tube.
19. Dendrites are numerous and do not extend far from the neuronal cell body or **soma,** axons may extend for several feet.
20. One agent of ventral specification is **sonic** hedgegog protein.
21. In frogs and chicks, **secondary** nueralation is seen in neural tube.
22. The cells at dorsal most portion of neural tube become **neural crest cells.**
23. About **33%** of identical twins have two complete and separate chorions
24. Mammals undergo holoblastic **rotational** cleavage.
25. In birds, **gravity** is critical in determining anterior posterior axis
26. Decidua becomes rich in blood vessels that will provide **oxygen** and nutrients to embryo.
27. Mammalian oocyte is released from ovary and swept by fimbriae into **the oviduct.**
28. Fertilization in chick egg occurs in **oviduct.**
29. Developmental decisions are usually **irreversible.**
30. All multicellular organisms arise by a slow process of progressive change called **development**.
31. Major class of cadherin includes **N- cadherin.**
32. Only **small %** of genome is expressed in partial cell.
33. Cortical are **granules** are membrane bound structure containing proteolytic enzymes.
34. Small subunits of MPF **cyclin-dependent kinase.**
35. It secretes a **cellulose** tunic (which gave the name "tunicates" to these creatures).
36. Inner surface cells thin 30×5um process **filopodia.**
37. β-catenin is Important for Inducing capacity of **micromeres.**
38. In fishes , First **12** divisions occur synchronously form a mouln of cells that sits an animal pole of large yolk cell.
39. Fertilization occurs in **ampulla** of oviduct a close, a region close to ovary.
40. In amphibians most of the xenopus most of tadpole is formed by **primary neurulation** and tail of neural tube formed by 2ndry neuralation.
41. Neural fold migrates to midline of embryo and fusing to form **neural tube** beneath the overlying ectoderm.

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42. **Tissues** form nervous system in the fly or in the frog.
43. The marginal zones to form clusters of neurons called **nuclei.**
44. Paracrine factors instruct **Myotome cells** to become muscles by inducing them to synthesize MyoD protein.
45. Sclerix is expressed in mesenchyme **from** the scleretome.
46. Acrosomal reaction in **sea urchins** is often highly specific to each species.
47. Cells are arrange their.. to creat new immunoglobin
48. In frogs and chicks, **secondary neurulation** is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae.
49. Regulation of gene expression by **several level.**
50. Which type of neuralation in fish **exclusively secondary**.
51. sonic hedgehocy is a type of **protein.**
52. **Blastomeres** have two parts of cell cycle.
53. Number of cortical granules Sea urchin **15000.**
54. How many strategies involve in cell commitment **2.**
55. Resting membrane potential of the egg is **70mv.**
56. One chemotactic molecule, a 14-amino acid peptide called **resact**.
57. Force for sperm propulsion is provided by **dynin.**
58. Regulative development arises from **Conditional Specification.**
59. Human brain contains **10^{12} glial cells.**
60. Fish have discoidal **telolicithal** pattern.
61. Process of osteogensis **2**
62. The mitotic spindle and contractile ring are **perpendicular** to each other.
63. Gastrulation in Snails is accomplished primarily by **epiboly.**
64. The first cell movement of fish gastrulation is the **epiboly** of the blastoderm.
65. This fetal organ the **Chorion** is derived primarily from embryonic trophoblast cells.
66. Secondary neurulation involves the making of a **medullary cord** and its subsequent hollowing into neural tube
67. The mantle zone cells differentiate into both **neurons** and **glia.**
68. The cerebral cortex in humans has **six** layers.
69. Paracrine factors such as **TGF** and FGF7 are important in normal skin development.
70. The **fifth** phase involves the invasion of the cartilage model by blood vessels.

71. Mode of commitment_____3___
72. strageries of commitment_____2___
73. Rana pipiens usually lays around __2500 eggs_____while the bullfrog, Rana catesbiana, can lay as many as 20,000
74. Multinucleated type of cell form._____syncytiotrophoblast
75. slow block, initiated by the intracellular release of_____ calcium ions
76. There are_____ two____ major modes of bone formation, or **osteogenesis**
77. Mutations of the ___*SOX9* gene____cause camptomelic dysplasia,
78. Human brain contains____**10^{12}** glial cells____
79. Telolecithal is cleavage is found in eggs of _____Birds and Fishes
80. ____Pax6 _____ plays a major role in eye formation
81. ___Pax1_____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
82. The eggs of insects have their yolk in the ____center_____ (i.e., they are centrolecithal),
83. Egg activates due to which ion more _____Ca_____
84. Which dermis give rise to brain _____Ectoderm____
85. Which dermis give rise to eye _____Ectoderm_____
86. Mod of commitment in insects _____syncytial specification____
87. Regulative development arises from_____ conditional specification
88. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___140 base pairs___ of DNA
89. the membrane potential of the sperm becomes ___more negative___ as potassium ions leave the sperm
90. Paracrine factors such as TGF- and FGF7 are important in normal _____skin development
91. In mouse ZP3 is attached to_____zona pellucid

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92. It is estimate ____egg is 25000 to 50000 mRNA. _____Sea Urchin
93. Fertilin protein is important for sperm-egg fusion in _____Mammals___
94. Fast block to polyspermy is initiated by _____Sodium___
95. Transition from fertilization to cleavage is caused by factor _____MPF
96. Eggs having sparse and equally yolk space are called _____isolethial___
97. Archenteron is used for _____primitive gut___
98. Spiral cleavage found in _____Snails____
99. Sonic hedgehog is a type of _____protein____

1. Which type of neurulation in fish ____secondary___
2. Nural crest divides functionally in domains ____4____
3. Myogenic bHLH proteins are specific for development of _____muscle cell___
4. __Zygote_____ which divide mitotically to produce all the cells of the body.
5. _____Tatratology_____ the study of birth defect.
6. Identical twins occur in roughly__0.25_ of human births.
7. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of ____identical twins.
8. Insect such as __drosophila____use all three mods of specification to commit their cells to particular fates.
9. The speed of metamorphosis is directly linked with environment ___pressures___
10. In acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.
11. The individuals of each species, however is divided into two mating types ___Plus and minus calamadomonas.
12. 140 base pairs of DNA _____nucleosome.
13. Viteline envelop contains at least _____eight____ different glycoproteins.
14. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by _____Lazzaro Spallanzani_____
15. The volume of the sea urchin is about ___200____ Pico liters
16. Hindbrain _____Rhombencephalon____
17. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of _____compaction____
18. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.
19. The neural form_____keel fish__
20. Spiral holoblastic clevage is characteristic of several animals groups, including **annelid** worms, some flatworms, and most mollusks (snails).
21. Diploblastic animals are those who have ectoderm and endoderm but no true mesoderm____(hydra)__
22. Even distribution of yolk in ovum_____isolecithal___
23. Size of bindin protein _____30,500-Da___
24. which has two parts in cell cycle options ____ Blastomere__
25. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the ___**germ ring**
26. Which forms spinal cord, hind brain and mid brain____ chordamesoderm.
27. This streak is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to koller sickle.
28. It will induce the uterine cells of the mammals to from the matrenal portion of the placenta, the__ Decidua.
29. Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or marginal belt).
30. The mitotic division of the nucleus is called ____karyokinesis___
31. Fishes, ___reptiles__, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.
32. The fast block, initiated by sodium influx into the cell in sea urchin.
33. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.
34. The rudiment of the central nervous system, is called neurulation.
35. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.
36. Fraternal twins are the result of two separate fertilization events.
37. The prechordal ____plate_____induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

38. The resulting larvae of snail is called _____glochidia____which resemble tiny bear traps.
39. D
40. Fertilization in the absence of sex occur through the __Binary Fission__
41. The macromeres divide meridionally, forming a tier of eight (8) cells
42. cells, derived from the micromeres, are called the primary mesenchyme.
43. Molluscus follow which type of cleavage ____spiral holoblastic cleavage
44. A mammalian blastula is called a blastocyst.
45. Which of the following undergo discoidal cleavage __fishes, reptiles and birds
46. The egg jelly factors that initiate the Acrosomal reaction in sea urchins are often highly specific to each species
47. The Metencephalon gives rise to the cerebellum
48. In primary neurulation, the original ectoderm is divided into three sets of cells
49. The growing region at the tip of the lip is called the chordoneural hinge
50. Neural Tube will form ___Brain and spinal cord
51. Each ICM (Inner Cell Mass) cell has the same potency
52. Variable cleavages produce no invariant fate assignments to cells
53. In early embryos of these insects, cell division is not complete
54. The neural crest cells originate at the dorsal most region of the neural tube
55. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs
56. the SOX9 gene, which encodes a DNAbinding protein
57. the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum
58. The second step is the alignment of the myoblasts together into chains
59. Cleavages sphere is known as __Blastula
60. Osteoclasts are probably derived from the same precursors as macrophage blood cells
61. charby desired to know the cause of birth defects
62. The direct conversion of mesenchymal tissue into bone is called intramembranous ossification
63. Those cells that remain integral components of the neural tube lining become ependymal cells
64. The original type of trophoblast cells constitute a layer called the cytotrophoblast
65. The major structural characteristic of avian, reptilian, & mammalian gastrulation is the primitive streak
66. The Blastomere cell cycle is governed by the synthesis and degradation of_____cyclin
67. Both have Anurans __and _. __Toad and frog
68. They have a notochord as larvae (chordates), but they lack vertebrae.
69. Acrosomal vesicles is derived from Golgi apparatus and contains __Enzymes__
70. The large subunit of MPF is _cyclin B
71. ZP3 is the specific glycoprotein in the mouse
72. MITF necessary for the production of __Pigment cells
73. Mutation of sox9 gene cause ____camptomelic dysplasia
74. Between the blastoderm and the yolk is a space called the sub-germinal cavity
75. The second step of gastrulation involves the involution of the mesoderm
76. The first cleavage furrow appears centrally in the blastodisc
77. Fertilization occurs in the ampulla of the oviduct
78. Mother provides fetus with __nutrients and oxygen
79. The first segregation of cells within the inner cell mass results in the formation of the hypoblast (sometimes called the primitive endoderm)
80. The fetal portion of placenta formed by __Chorion__
81. Paracrine factor instruct myotome cells to become__ Muscles
82. When myoblasts stop dividing, secrete fibronectin

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83. Retinoic Acid comes from **Somites**
84. Hox genes pattern the anterior-posterior axis and help to specify positions along that **axis**
85. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and **hypothalamic** brain regions
86. Endochondral ossification involves the formation of **cartilage**
87. J.Hammerling exchanged nuclei b/w two **morphologically** distinct species
88. When plus individuals initiate fusion by extending a fertilization tube. Which contacts & fuses with a specific site on the **minus** individual.
89. Promoter is **enhancer** which is thought to regulate transcription.
90. Identical twins share a common **Chorion**
91. In Birds, **pH** differences appear crucial for distinguishing dorsal from ventral
92. Which regulate function of Promoter Enhancer
93. The space between two neurons, where transmitter are released **synapse**
94. integral components of the neural tube lining become **ependymal**
95. Mantle cells differentiate into **Gray matter**
96. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area **opaca.**
97. During fertilization, the **actin** molecules fuse to form the long microfilaments.
98. A condition in which cell lack the instructions for development is **naive**
1. metamorphosis is initiated by hormones from the tadpole's **thyroid gland**
2. Brain has **100** billion neurons
3. Release of Ca from intracellular storage controlled by dye **Aequorin**
4. Nervous system develop by **Neural Tube**
5. Allows the mRNA to exit the nucleus **poly a tail**
6. Contact and ___between sperm and egg **Recognition**
7. The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by ____in the late 1700s. **Lazzaro Spallanzani**
8. The early embryonic cells need a supply of energy and amino acids which is provided by **yolk** proteins in the egg.
9. The depression is called the primitive **groove**

ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make Globins
2. how many strategies for developmental commitment 2
3. Ribosomes produce in sea urchin 10^{12}
4. Resact is isolated from egg jelly of sea urchin
5. ___is a discipline that studies embryonic & other developmental processes Developmental Biology
6. 0nly small %age of genome expressed in particular cell
7. Digestive & respiratory system formed by Endoderm
8. Progesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are ..... specification syncytial
10. In Chlamydomonas haploid chromosome has.....copy one
11. Modes of commitments are..... 3
12. After fertilization developmental process proceed by Cleavage
13. Ca ion responsible for cortical granules stores in Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration Na (Sodium)
15. There are ___approaches of anatomy 4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. *Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. Egg
19. Resact diffuse readily in sea water.
20. The requirement of capacitance varies from species to Species.

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21. The division of embryo into 3 _____Germ layers
22. Cell potency____≥____fate
23. Zona pellucida_____ also initiate the acrosomal reaction after sperm bound to it.
24. Mammalian egg is surrounded by layer called. _____Cumulus
25. The volume of the sea urchin is about ___200_____ Picoliters.
26. The sperm travel by the amoeboid motion of _____ lamellipodial extensions___of the cell membrane.
27. The sex and ___Reproduction_____ process included into fertilization.
28. The term spermatozoa means.. _____Sperm Animals
29. In 1824_____J. L. Prevost and J. B. Dumasclaimed_____ sperm is not parasite.
30. 4 cadherin classes are found in _____Vertebrates_____ embryos. .
31. The second major group of protostomes are the _____Lophotrochozoa___
32. Sperm discovered in_____1678____
33. Transcription factors have ____three____ major domains
34. How many Amphibian oocytes produces ribosomes as_____10^{12}
35. Enhancers can also be used to inhibit _____transcription____
36. In deutrosomes the mouth opening would formed after... opening _____Anal___
37. Germs cell formed in...._____specialized portion of egg cytoplasm
38. Syncytial specification found in _____insects____
39. Third anatomical approach to developmental biology is _____Teratology____
40. The mature sea urchin is metabolically sluggish and activated by. _____Sperm___
41. The resting membrane potential is.....mv. _____70mV
42. Fusion is a ___Active___process that is activated by fusogenic proteins.
43. Axoneme s formed by. _____Microtubules___
44. Eukaryotic genes are contained within a complex of DNA and protein called___Chromatin____
45. The negative enhancers are called____Silencers____
46. The ability of embryonic cells to change their fates is called.....____Regulation____
47. ____Pax6____transcription factor found in mammalian eye, nervous system and pancreas.
48. E- Cadherin is expressed on all early mammalian embryonic cells.
49. Annelids, molluscs and flatworms belong to..... ___ Lophotrochozoa
50. Potency____ of cell decrease if cell is committed.
51. Fertilization in Rana Pipiens usually lasts for about __3 (three)_ years.
52. An uncommitted cell can be described as ____Naive____
53. Lying immediately beneath the plasma membrane of the egg is a thin ____Shell___
54. Acroome can be considered as modified secretory_____ **vesicles**
55. Most marine organisms release their ____gametes____ into the environment
56. The fast block method is used to prevent_____Polyspermy___
57. Autonomous specification was first demonstrated in_____1887
58. In mouse ZP3 is attached to_____zona pellucid
59. TAFs used in_____transcription factor activity
60. in slow block method which is cortical granule protein_____ hyalin and peroxidase
61. Egg shell formed _____weight of egg._____9 – 12%
62. potency is ___intrinsic___ property
63. cytoplasm that contains many nuclei called _____ syncytium
64. Ectoderm forms..... __ epidermis, nervous system, and pigment cells.
65. according to dutch sperm are ...___ parasitic animals (spermatozoa)
66. Acrosome & nucleus constitute the head of the sperm.
67. cortical granules are _____membrane bound structures
68. Acrosomal reaction has 2 (two) component
69. Acrosome-reacted sperm transfer their binding from ZP3 to the adjacent ZP2 molecules
70. Ca & bicarbonate ions maybe critical in activating cAMP
71. Peroxidase enzyme hardens fertilization envelope
72. The Fast block to polyspermy is achieved by changing the electric potential
73. P-cadherin stands for _____Placental Cadherin
74. Most transcription factor can bind to specific DNA sequences
75. MITF protein necessary for production of pigments cell and their pigments.
76. Outside of plasma membrane is the vitelline envelope.
77. Mouse ___**Fertilin**___ is localized to posterior plasma membrane of the sperm **head**
78. Finally the fourth cortical granules protein ____hyaline____ forms a coating around egg.

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79. The cleavage stages are called.____Blastomere
80. Enhancers are.....DNA Sequences
81. The mitotic spindle and contractile ring are perpendicular to each other
82. J.Whittaker stained blastomeres for the presence of the enzyme acetyl cholinesterase
83. Protocadherins are involved in separating notochord from mesoderm in __Xenopus gastrulation
84. Pax6 factor needed for mammalian eye
85. Which stands comparative embryology is ____First
86. Cell type specification precedes any large-scale embryonic cell migration
87. Series of extremely rapid divisions in zygote is ___Cleavage
88. In protosome, mouth is formed first than Deuterostomes: forms.
89. Northern blots, in situ hybridization, & the PCR can show which cells are transcribing particular genes.
90. Coding regions of DNA are called __Exon
91. Na and __K__ Are important in fast block polyspermy.
92. The force for sperm propulsion is provided by __dynein
93. Spallenazani demonstrated that filtered __toad__ semen devoid of sperm would not fertilize egg.
94. Mesoderm generates the____KIDNEY.
95. Sperms are attracted towards egg of their species by CHEMOTAXIS.
96. Mathematical modeling seeks to desire developmental phenomenon in terms of EQUATIONS.
97. A region that is not translated in proteins is called UNTRANSLATED region.
98. Invariant cleavage produce the ____SAME____ lineages in embryo
99. Germ layers are formed as a result of ___GASTRULATION
100. RNA polymerase binds with PROMOTOR in start of transcription
101. The region of globular actin molecule lies between nuclear and acrosomal vesicle in ____SEA URCHIN
102. Which technique is used in situ hybridization ___ Northern blots
103. Conditional specification involves interaction with RELATIVE POSITIONS cells
104. Particular proteins or carbohydrate on the sperm surface ___LOST___ during capacitation
105. ___LOSS____of hydrogen ion causes pH to rise.
106. Peroxidase enzyme hardens the fertilization envelope by crosslinking TYROSINE residues
107. FERTILIZATION transmits genes from parents to offspring.
108. Regulation of gene expression accomplished by SELECTIVE MESSENGER RNA TRANSLATION
109. Development contact genotype and PHENOTYPE
110. Nucleosome is composed of OCTAMER of histone protein.

1. Knowledge of gene activity in humans can be obtained by candidate GENE MAPPING & POSITIONAL CLONING
2. Anatomical abnormalities are caused by MUTANT GENES
3. All multicellular organisms arise by a slow process of progressive change called DEVELOPMNET
4. Animals development is the division of embryo into three layers Ectoderm, Mesoderm and___ENDODERM
5. The ions which are responsible for increasing granule reactions_____Ca (Calcium)
6. The zona pellucida is a fglycoprotein matrix, which is synthesized and secreted by the growing oocyte
7. In several other species, certain regions of the membrane are specialized for sperm recognition & fusion
8. An intrinsic program has been activated within the cell that causes it to follow a particular pathway of development
9. Activation of egg metabolism to start development
10. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates
11. Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in Acetabularia.
12. Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly
13. Bacteria are able to transmit genes from one individual to another by means of sex pili
10. +ve and – ve mating take place *Chlamydomonas*

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Q1: Number of cortical granules Sea urchin **15000**

Q2: Fertilin protein is essential for sperm-egg membrabe fusion found in **Mammales**

Q3: Which of the following uses all 3 modes of specification? **Drosophila**

Q4: The 1st evidence suggesting the importance of sperm in reproduction shown by **Lazzaro Spallanzani**

Q5: Transcription factors have ---------- major domains **3**

Q6: Nuleosoms contain approximately ----------base pairs of DNA **140**

Q7: Cadherins are ---------ion dependent adhesion molecules **Calcium**

Q8: Chordates are included in **Deuterostomes**

Q9: In *Acetabularia* development is controlled at --------- levels

both the transcriptional and translational

Q10: Estrogen is a hormone that can instruct which organ of the body to secrete the yolk proteins **Liver**

Q11: Metamorphosis is initiated by hormones from the tadpole's

Thyroid Gland

Q12: J.R.Whitakar worked on blastomere to check out which of the following enzyme **Acetylecholinesterase**

Q13: A condition in which cell lacks the instructions for development is **Naïve**

Q14: Which of the following divide mitotically to produce all the cells of the body

Zygote

Q15: How many strategies involve in cell commitment **2**

Q16: Which of the following Zona Pellucida of the mouse sperm binds **ZP3**

Q17: The resting membrane potential of the egg is **70mv**

Q18: Calcium and bicarbonate ions may be critical in activating **cAMP**

Q19: One chemotactic molecule, a 14-amino acid peptide called **Resact**

Q20: The force for sperm propulsion is provided by **Dyenin**

1) J.R Whittaker stained blastomere in the presence of enzyme**.....(Acetylcholinesterase)**
2) The nuclei region containing high amount bicoid will be instructed to activate those gene necessary for producing...(**head)**
3) A cytoplasm that contain many nuclei is called**...(Syncytium)**
4) Potency is...(**intrinsic property)**
5) Autonomial abnormalities may be caused by ....(**mutant genes)**